Mehmet Güzel

Software Engineer • Backend & DevOps

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Languages: English (B2) | Turkish (native)

Summary

Backend engineer (Java/Spring Boot, Python/FastAPI) with hands-on experience in event-driven microservices, a published CVE from an application security internship, and CI/CD pipelines deployed to AWS. Software Engineering graduate, June 2026.

Experience

Backend Developer Intern

August 2026 – September 2026

Codexist — *Istanbul, Türkiye*

- Designed the architecture of a **GEO (Generative Engine Optimization)** system that measures how visible companies are in LLM answers compared to competitors, running 126 prompts per day across 4 LLM providers and reporting results with a statistical margin of error.
- Built daily automated crawl-and-analysis jobs on **Celery** and **Redis**; implemented **sliding-window** workers that reduce the measurement error of repeated LLM queries.
- Exposed the analysis pipeline as **FastAPI** services backed by **PostgreSQL**, connected them to a **TypeScript/React** frontend, and packaged the stack with **Docker Compose**.

Application Security Intern

June 2024 – July 2025

Privia Security — *Istanbul, Türkiye*

- Discovered a stored XSS vulnerability in a commercial product and reported it through responsible disclosure; it was officially assigned **CVE-2025-1301**.
- Analyzed production web applications at the code and API level and took part in web application penetration tests.
- Built internal **Python** tools that automated repetitive manual security checks for the team.
- Worked with development teams on remediation by proposing code-level fixes for the vulnerabilities found.

Projects

STOP — Event-Driven Microservice Platform *(B.Sc. graduation project)* mehmettguzell/STOP

Java 21 · Spring Boot · Apache Kafka · PostgreSQL · Redis · Docker · GitHub Actions · AWS

- Built the backend of a football match organization platform as 5 independent **Spring Boot** services (API Gateway, Identity, Match, Communication, Notification), each with its own **PostgreSQL** database and **Flyway** migrations.
- Connected services with **Kafka** events across 20+ topics (matches, participation, waiting lists, chat); implemented a FIFO waiting list that admits the next player automatically and live chat over **WebSocket (STOMP)**.
- Secured the system with **JWT** verified by each service against its own RSA public key, plus Redis-backed rate limiting (**Bucket4j**) and a **Resilience4j** circuit breaker at the API Gateway.
- Set up **GitHub Actions** CI/CD that builds only changed services and pushes images to GHCR tagged with the Git SHA; deploys to **AWS EC2** via SSM Run Command (no SSH), with secrets from Secrets Manager through an IAM instance role, automatic rollback on failed health checks, and Prometheus/Actuator metrics.

DevOps Todo App — End-to-End CI/CD Pipeline mehmettguzell/Devops-Todo-App

GitHub Actions · Docker · GHCR · AWS EC2 · IAM · SSM · OIDC · Python · TypeScript

- Built a **GitHub Actions** pipeline for a FastAPI backend and TypeScript frontend covering test, release, build, and deploy; `paths-filter` and a dynamic build matrix test and build only the changed services.
- Deployed to **AWS EC2** with no static credentials: CI assumes a temporary IAM role via **OIDC**, deploys with SSM Run Command, and the server pulls production configuration from SSM Parameter Store with its own IAM role.
- Wrote multi-stage **Dockerfiles** and a Docker Compose setup with health checks, startup ordering, and resource limits; the deploy fails the pipeline if services do not become healthy. Wrote the EC2 bootstrap script and automated versioning with release-please.

Event-Driven Order Management Platform — Distributed Transactions Event-Driven-Order-Management-Platform

Python · Django REST Framework · RabbitMQ · PostgreSQL · JWT · Nginx · Docker Compose

- Built a 5-service commerce backend (user, product, order, inventory, payment) with asynchronous stock and payment processing over **RabbitMQ**, the **Saga pattern** with compensating transactions for consistency, and row-level locking for concurrent stock updates.

- Provided routing, rate limiting, and JWT authentication through an **Nginx** API gateway; each service owns its database.

Education

Maltepe University September 2021 – June 2026
B.Sc. in Software Engineering (English-medium) *Istanbul, Türkiye*

Technical Skills

Languages: Java, Python, TypeScript, Go, SQL, Bash

Backend: Spring Boot (Security, Data JPA, Cloud Gateway), FastAPI, Django REST Framework, REST/OpenAPI, Microservices, Event-Driven Architecture

Data & Messaging: PostgreSQL, MySQL, MongoDB, Redis, Apache Kafka, RabbitMQ, Celery, Flyway

DevOps & Cloud: Docker, Docker Compose, GitHub Actions, Amazon Web Services (AWS: EC2, IAM, SSM), Nginx, Linux, Prometheus

Security & Testing: OWASP Top 10, JWT, OAuth2/OIDC, JUnit, Testcontainers, pytest

Achievements & Certifications

Competition: **Future Commerce AIdeathon** — Ministry of Trade, Türkiye E-Commerce Week; AI-based dead stock optimization

Certifications: CSD Java Programming Course · BTK Akademi Cyber Security Camp